



The Good Morning Bed

An IoT-Based Wake-Up & Sleeping System

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The Problem with Alarms

Sudden Shock

Loud alarms interrupt sleep cycles, raising stress hormones instantly

Groggy Start

Waking too fast leaves you tired and stressed all day

No Awareness

Traditional alarms don't know if you're even in bed

The Solution: A Smart Bedroom



Core Goals



Replace loud alarms with gentle, step-by-step room changes

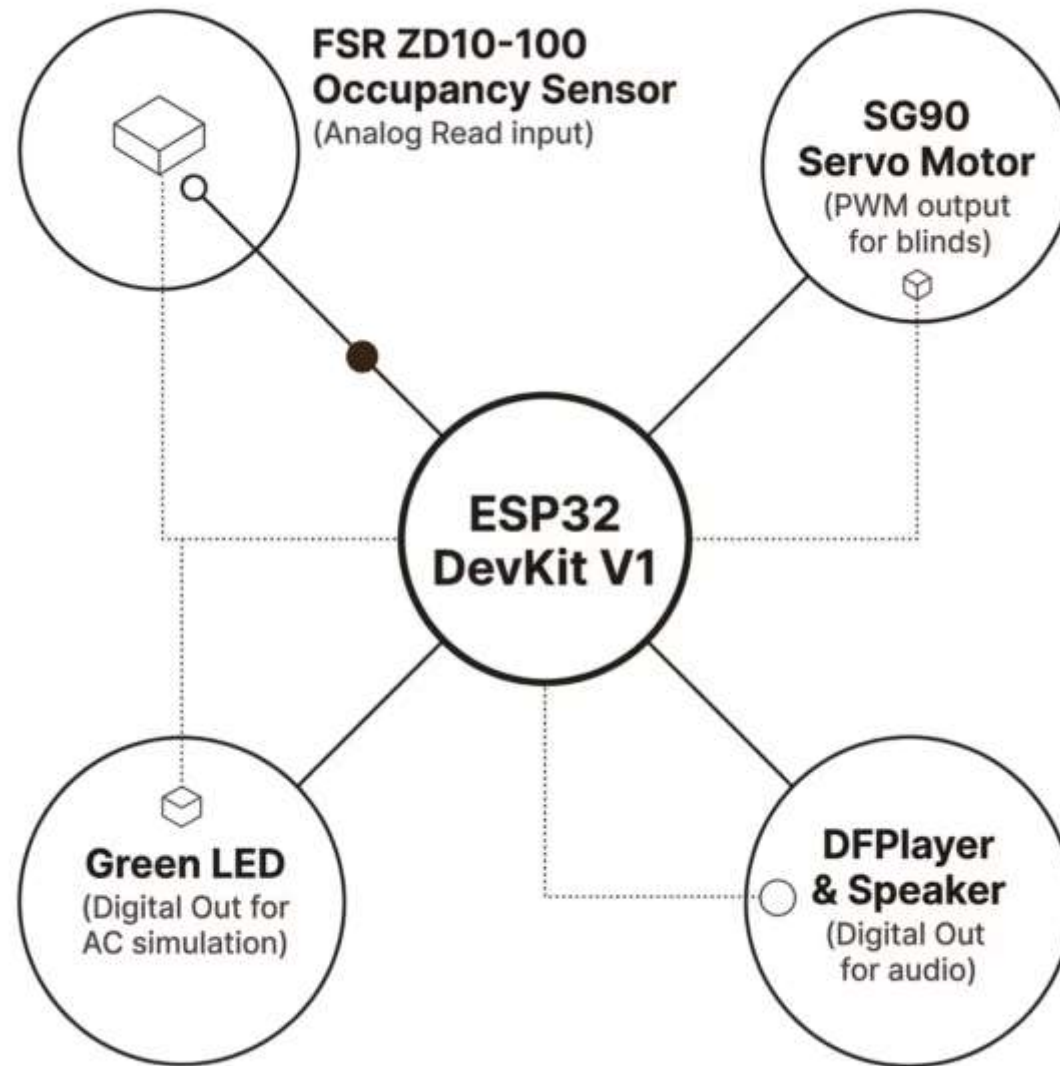


Auto-control climate via Green LED (AC simulation)



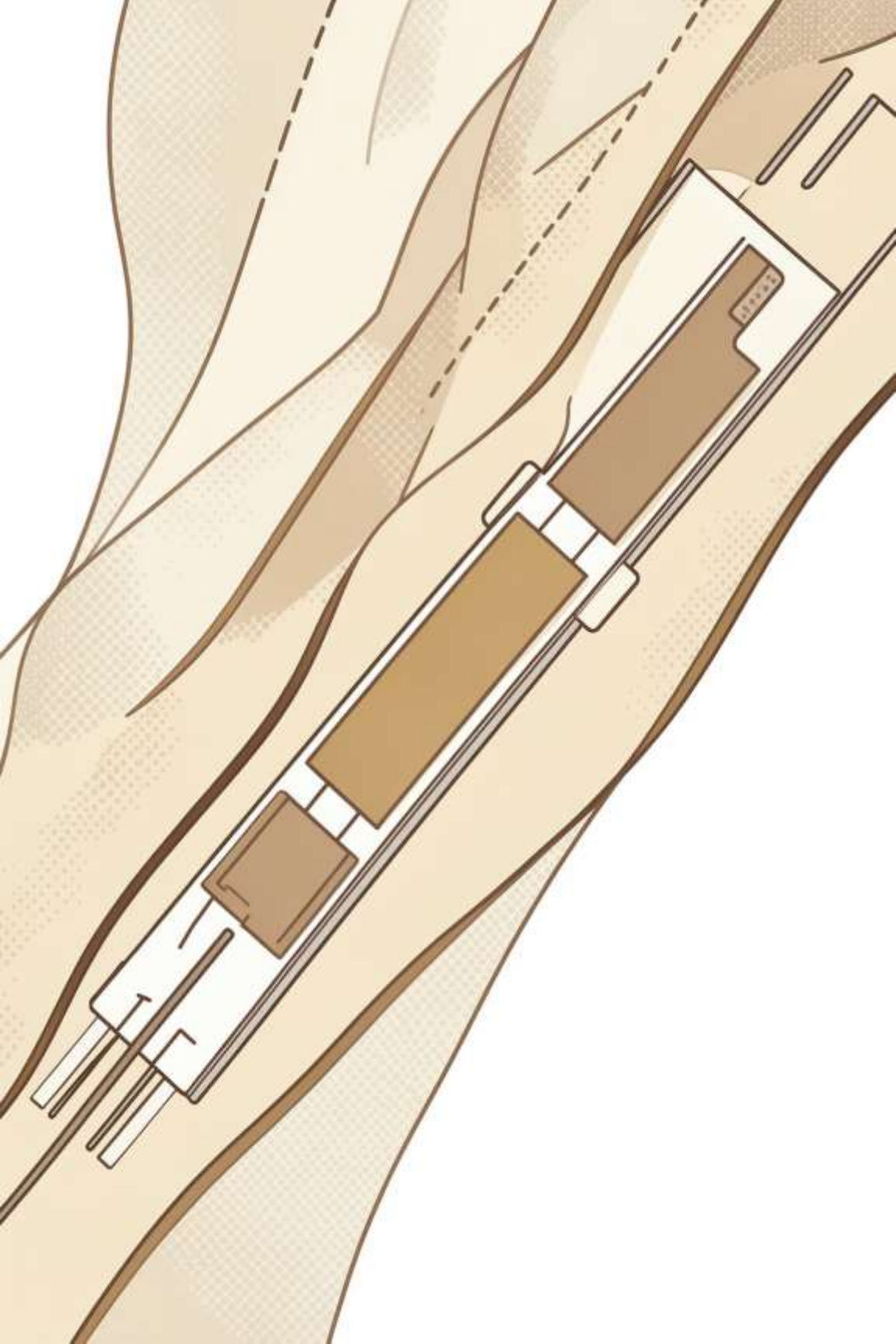
Only trigger wake-up if occupancy is physically detected

System Architecture



The ESP32 DevKit V1 is the central brain — reading sensors and commanding all actuators in a continuous loop.

i Built-in Wi-Fi & Bluetooth allow future smart home integration.



Occupancy Sensor: ZD10-100 FSR

How It Works

Force compresses conductive layers → resistance drops → voltage signal rises to ESP32

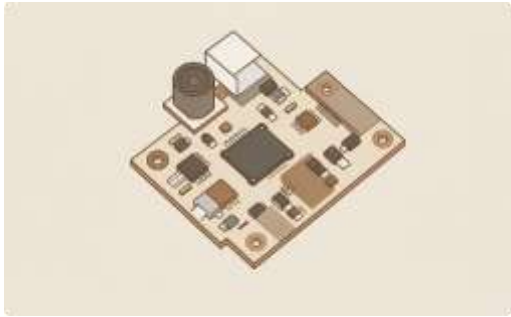
Smart Threshold

Programmed cut-off distinguishes a person from a heavy object

Signal Path

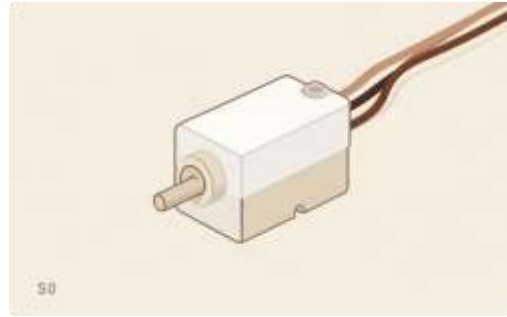
3.3V → FSR → 10kΩ pull-down → V_{out} to ADC

Audio & Physical Actuators



DFPlayer Mini + Speaker

Plays white noise to sleep, then gentle nature sounds that slowly escalate in volume



SG90 Servo Motor

Rotates 90° to simulate opening window blinds — natural light stops sleep hormones



Green LED (AC Sim)

LED ON = cool room for sleep. LED OFF during wake-up = room warms, signaling body to rise

The Logical Flow

FSR Detects
Occupancy

Enter Sleep
Mode

Enter Wake
Mode

Scheduled
Wake Trigger



The ESP32 runs a continuous loop, using real-time FSR data to instantly switch between Sleep Mode and Wake Mode.

The Awakening Algorithm

01

Occupancy Check

Bed empty? Alarm cancelled. No wasted power.

02

Climate Shift

LED OFF — room warms slightly

03

Light Actuation

Servo opens blinds — sunlight enters

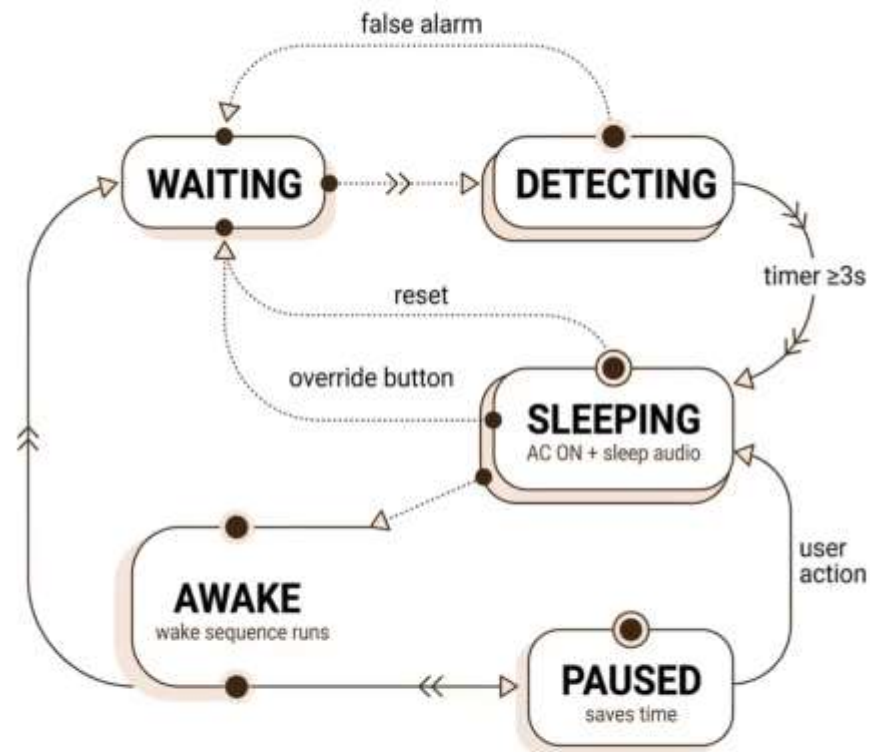
04

Sound Escalation

Nature audio starts low, rises until you rise



System State Machine



A finite state machine governs all behavior — moving between five states based on time thresholds and sensor input.

WAITING

DETECTING

SLEEPING

AWAKE

PAUSED

Conclusion & Future Vision

Today

FSR + ESP32 + Servo + DFPlayer + LED = a gentle, biology-aware wake-up prototype

Tomorrow

Add machine learning to track sleep cycles and wake you during your lightest sleep phase



Thank You